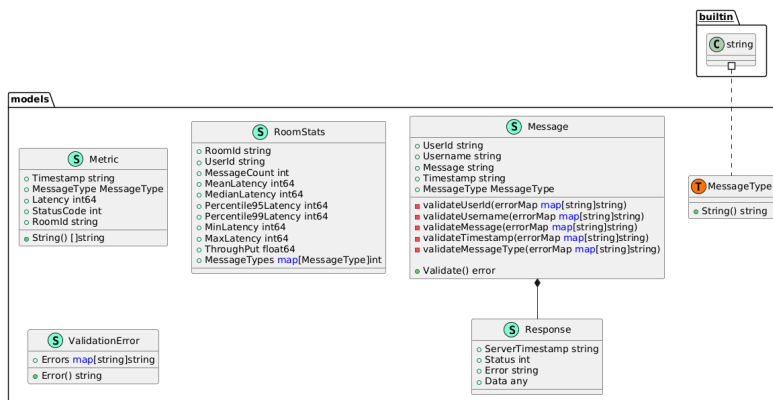
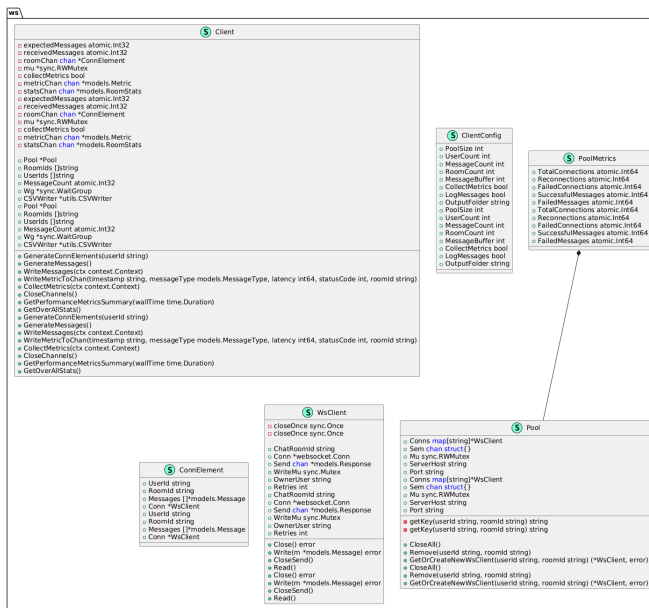
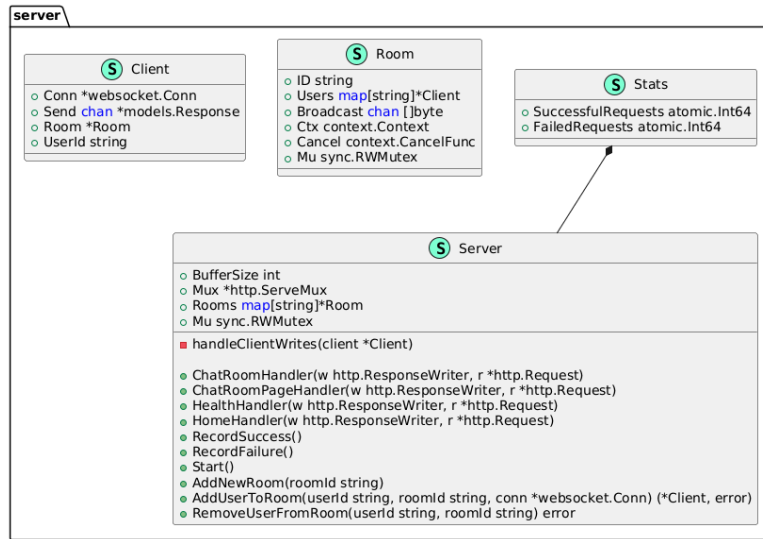


1. **Github URL:** <https://github.khoury.northeastern.edu/supriyakotturu/chat-flow>
2. **Class Diagram:**



Server threading :

<https://excalidraw.com/#json=vHp4Ci1Ue1VMGajPUVby2,RWQTy7On3b5l1buUmXJ2g>

Client threading:

- Fan out model [part-1]
https://excalidraw.com/#json=u0nf4FPjrmAMHEPUqXN3s,ZpCUDLTlc_5Obli9EIzINA
- Pipeline model [part-2]
<https://excalidraw.com/#json=lwcm5yhSsf2NXjCMgZsVz,5F2gHHvub0roKjc46ghgag>

Little's law: estimated throughput = concurrency / response time

Pipeline - for m users, n rooms, k messages

- Generator goroutine - n [produce n connElements]
 - Writer goroutines - m * n [send messages to server]
 - Reader goroutine - m * n [read messages sent from server & compute stats]
 - WsClient.Read goroutines - m * n [read from ws and push to readChan]
 - Metric collector - 1 [collect metrics from metricChan & write to CSV]
-
- Response time for one user to send one message to one room = 30ms

```
→ chat-flow git:(main) go run ./client-part2/main.go
Starting pipeline load test...

=== Performance Metrics ===
Total runtime (wall time): 0.1s
Successful messages sent: 3
Failed messages: 0
Overall throughput: 38.7 msg/s
Total connections: 1
Reconnections: 0
Failed connections: 0
Mean Latency across all rooms: 30ms
Median Latency across all rooms: 30ms
95th Percentile Latency across all rooms: 30ms
99th Percentile Latency across all rooms: 30ms
Min Latency across all rooms: 30ms
Max Latency across all rooms: 30ms
Median Throughput across all rooms: 91.22 msg/s
Throughput across rooms:

Room 7 | users: 1 | throughput: 91.2 msg/s | mean latency: 30ms | median latency: 30ms
Message Types: JOIN: 1 TEXT: 1 LEAVE: 1
→ chat-flow git:(main)
```

Expected throughput = $3(m*n) + n + 1 / 30ms$

For 1 user ->

Goroutines = $3(1*1) + 1 + 1 = 5$

$3(1*1) + 1 + 1 / 30ms = 5 / 30ms = 5 / 0.030 = 166.7 \text{ msg/s}$

For 500k users [m=50, n = 20, k = 500, pool = 1000] ->

Mean latency = 1166ms

Goroutines = $3(50*20) + 20 + 1 / 0.030 = 3021 / 1.166 = 2590.9 \text{ msg/s}$

Sai Supriya Kotturu

Chatflow - assignment-1

Test Results:

Part-1 - 6.5s

```
=== Performance Metrics - Warm up ===
Total runtime (wall time): 6.5s
Successful messages sent: 320640
Failed messages: 0
Overall throughput: 49237.7 msg/s
Total connections: 320
Reconnections: 0
Failed connections: 0
→ chat-flow git:(main) 🌸 🍄 🍌 🍌
```

Part-2 - 8.5s

```
=== Performance Metrics ===
Total runtime (wall time): 8.5s
Successful messages sent: 502000
Failed messages: 0
Overall throughput: 58943.8 msg/s
Total connections: 1000
Reconnections: 0
Failed connections: 0
Mean latency across all rooms: 1166ms
Median latency across all rooms: 1072ms
95th Percentile Latency across all rooms: 2481ms
99th Percentile Latency across all rooms: 3380ms
Min latency across all rooms: 96ms
Max latency across all rooms: 3948ms
Median Throughput across all rooms: 203.34 msg/s
Throughput across rooms:

Room 6 | users: 50 | throughput: 10354.5 msg/s | mean latency: 873ms | median latency: 787ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 14 | users: 50 | throughput: 7161.2 msg/s | mean latency: 1868ms | median latency: 1683ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 17 | users: 50 | throughput: 7739.1 msg/s | mean latency: 1682ms | median latency: 1569ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 7 | users: 50 | throughput: 8417.8 msg/s | mean latency: 1561ms | median latency: 1361ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 18 | users: 50 | throughput: 10206.8 msg/s | mean latency: 1308ms | median latency: 1271ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 15 | users: 50 | throughput: 23842.5 msg/s | mean latency: 804ms | median latency: 826ms
Message Types: TEXT: 25000 LEAVE: 50 JOIN: 50

Room 4 | users: 50 | throughput: 18311.3 msg/s | mean latency: 710ms | median latency: 555ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 16 | users: 50 | throughput: 7464.9 msg/s | mean latency: 1796ms | median latency: 1774ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 19 | users: 50 | throughput: 13446.9 msg/s | mean latency: 1208ms | median latency: 1230ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 1 | users: 50 | throughput: 52494.9 msg/s | mean latency: 306ms | median latency: 277ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 3 | users: 50 | throughput: 7326.8 msg/s | mean latency: 1226ms | median latency: 1052ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50
```

```
Room 16 | users: 50 | throughput: 7464.9 msg/s | mean latency: 1796ms | median latency: 1774ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 19 | users: 50 | throughput: 13446.9 msg/s | mean latency: 1208ms | median latency: 1230ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 1 | users: 50 | throughput: 52494.9 msg/s | mean latency: 306ms | median latency: 277ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 3 | users: 50 | throughput: 7326.8 msg/s | mean latency: 1226ms | median latency: 1052ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 10 | users: 50 | throughput: 7967.9 msg/s | mean latency: 1521ms | median latency: 1430ms
Message Types: LEAVE: 50 JOIN: 50 TEXT: 25000

Room 9 | users: 50 | throughput: 10063.7 msg/s | mean latency: 1194ms | median latency: 1174ms
Message Types: LEAVE: 50 JOIN: 50 TEXT: 25000

Room 13 | users: 50 | throughput: 11169.3 msg/s | mean latency: 1287ms | median latency: 1198ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 12 | users: 50 | throughput: 31568.6 msg/s | mean latency: 648ms | median latency: 667ms
Message Types: LEAVE: 50 JOIN: 50 TEXT: 25000

Room 2 | users: 50 | throughput: 81532.9 msg/s | mean latency: 204ms | median latency: 200ms
Message Types: TEXT: 25000 LEAVE: 50 JOIN: 50

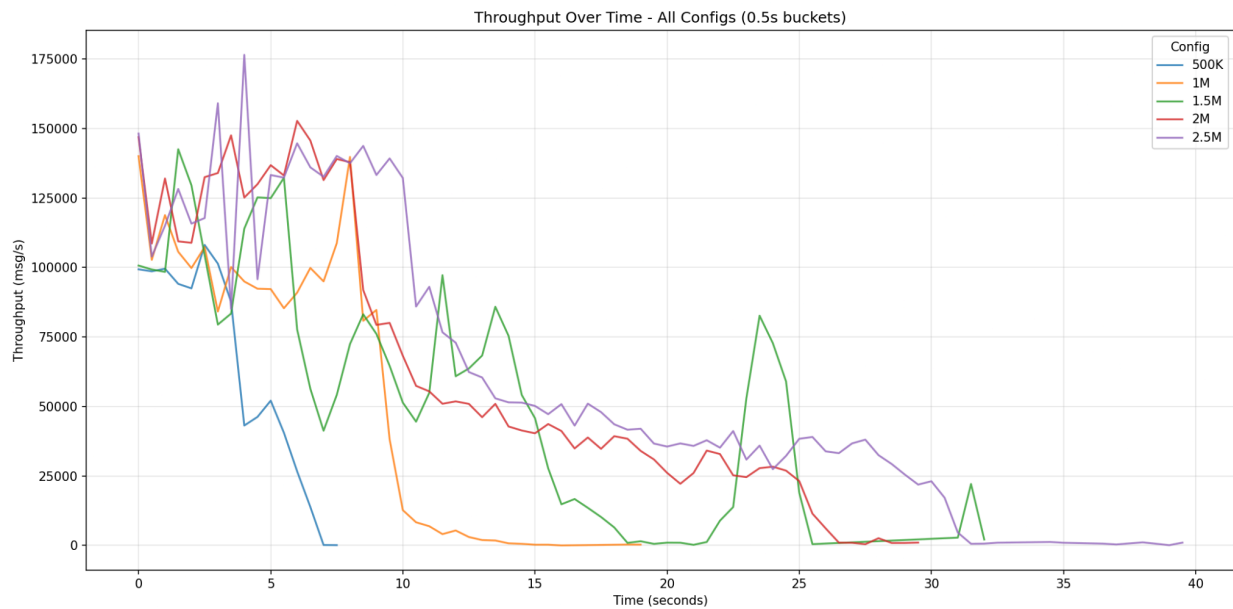
Room 20 | users: 50 | throughput: 7031.6 msg/s | mean latency: 1451ms | median latency: 1433ms
Message Types: TEXT: 25000 LEAVE: 50 JOIN: 50

Room 11 | users: 50 | throughput: 8662.1 msg/s | mean latency: 1368ms | median latency: 1270ms
Message Types: TEXT: 25000 LEAVE: 50 JOIN: 50

Room 8 | users: 50 | throughput: 9529.4 msg/s | mean latency: 1261ms | median latency: 1182ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50

Room 5 | users: 50 | throughput: 17952.8 msg/s | mean latency: 1052ms | median latency: 1078ms
Message Types: JOIN: 50 TEXT: 25000 LEAVE: 50
→ chat-flow git:(main) 🌸 🍄 🍌 🍌
```

Performance Analysis chart:



EC2 deployment:

```

~ ~ ~      \###|
~ ~ ~      \###|
~ ~ ~      \###|
~ ~ ~      \|/  https://aws.amazon.com/linux/amazon-linux-2023
~ ~ ~      V~'  '->
~ ~ ~      /
~ ~ ~  _  /
~ ~ ~  _  /
~ ~ ~  _  /m/'

Last login: Fri Feb 17 05:38:58 2026 from 73.225.128.242
[ec2-user@ip-172-31-23-171 ~]$ cd chatflow/
[ec2-user@ip-172-31-23-171 chatflow]$ ls
bin
[ec2-user@ip-172-31-23-171 chatflow]$ ./bin/server/chat-server
2026/02/13 20:29:26 ChatRoomHandler called for path: /chat/7
2026/02/13 20:29:26 User 1 left room 7
2026/02/13 20:30:18 ChatRoomHandler called for path: /chat/7
2026/02/13 20:30:18 User 1 left room 7

```

2026/02/13	20:59:06	User 47	left	room 17
2026/02/13	20:59:06	User 51	left	room 6
2026/02/13	20:59:06	User 74	left	room 15
2026/02/13	20:59:06	User 84	left	room 6
2026/02/13	20:59:06	User 14	left	room 15
2026/02/13	20:59:06	User 27	left	room 17
2026/02/13	20:59:06	User 37	left	room 17
2026/02/13	20:59:06	User 35	left	room 12
2026/02/13	20:59:06	User 95	left	room 2
2026/02/13	20:59:06	User 54	left	room 15
2026/02/13	20:59:06	User 32	left	room 2
2026/02/13	20:59:06	User 49	left	room 19
2026/02/13	20:59:06	User 36	left	room 12
2026/02/13	20:59:06	User 44	left	room 15
2026/02/13	20:59:06	User 100	left	room 9
2026/02/13	20:59:06	User 10	left	room 5
2026/02/13	20:59:06	User 64	left	room 9
2026/02/13	20:59:06	User 81	left	room 5
2026/02/13	20:59:06	User 56	left	room 19
2026/02/13	20:59:06	User 57	left	room 13
2026/02/13	20:59:06	User 61	left	room 17
2026/02/13	20:59:06	User 45	left	room 12
2026/02/13	20:59:06	User 64	left	room 5